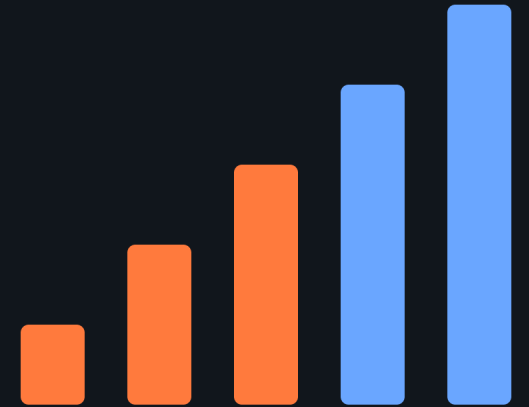


[EvoNexus · Oct 8th]

Scale Software Development with AI Sandboxes

From laptop steering to agent swarms

Mathew Mathew · CTO / Claramap.com]





LET'S CONNECT

Mathew Mathew

CTO & Founder, Claramap

Claramap helps companies design, build and run **agentic systems** in production.

20+ years as a software engineer and data scientist at IBM Cloud, Amdocs, AT&T, SolarWinds and Discover.



SCAN TO CONNECT

[linkedin.com/in/mathewma](https://www.linkedin.com/in/mathewma)

github.com/mathaix/fife

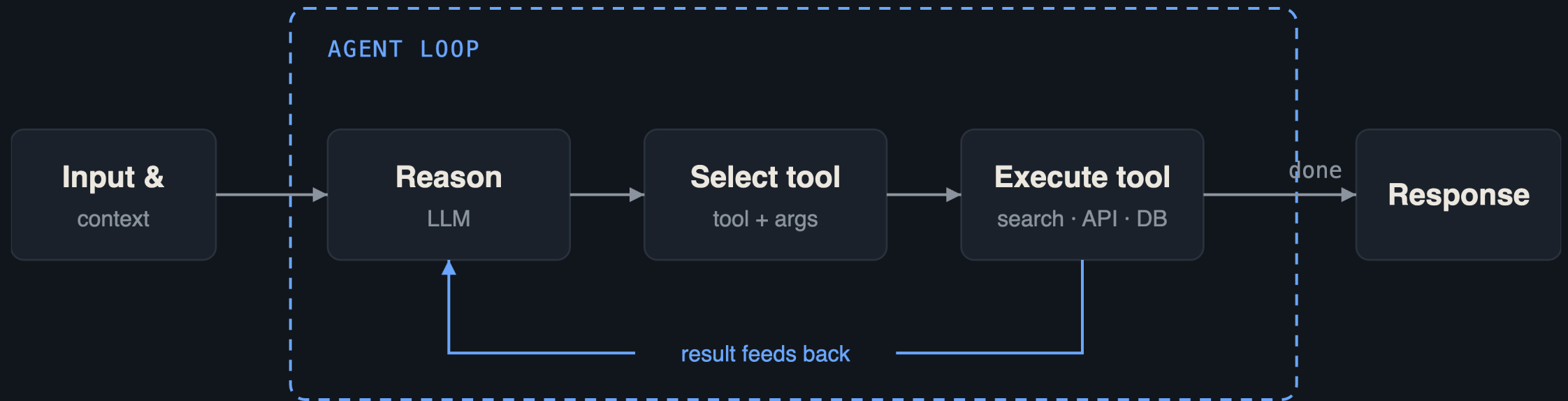
github.com/mathaix/signals

BEFORE WE START

Where are you on the ladder?

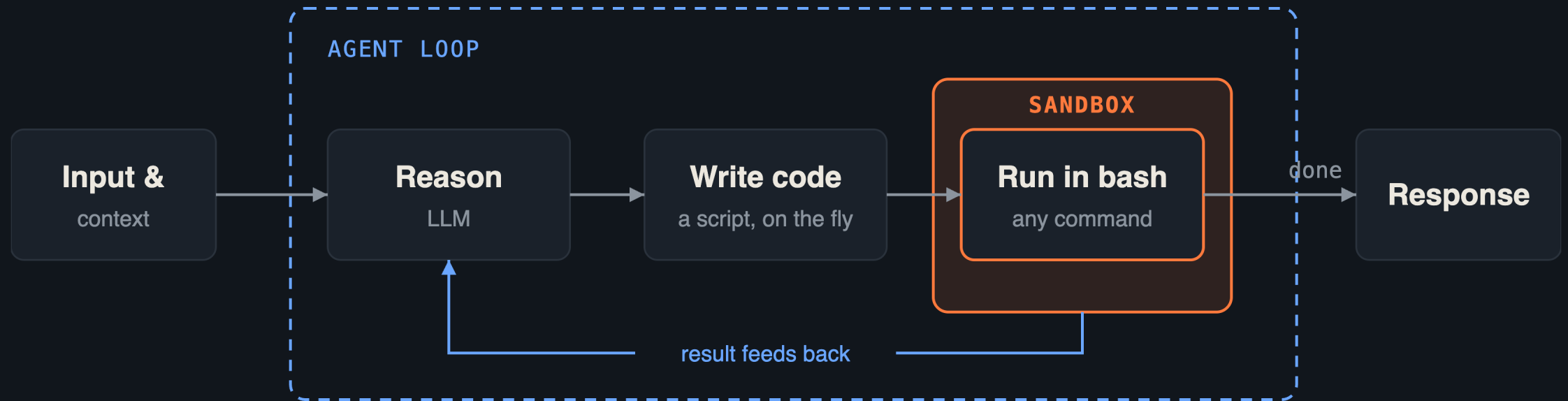
5	Agent swarm	Many agents, one problem
4	Agent factory	Tickets in, PRs out
▲ SANDBOXES LAPTOP ▼		
3	Parallel local agents	Worktrees and YOLO mode
2	Local agent	An agent in your terminal
1	Autocomplete	AI suggests, you do the rest

The agent loop



Every action is a tool you defined. You know exactly what it can do.

The coding agent



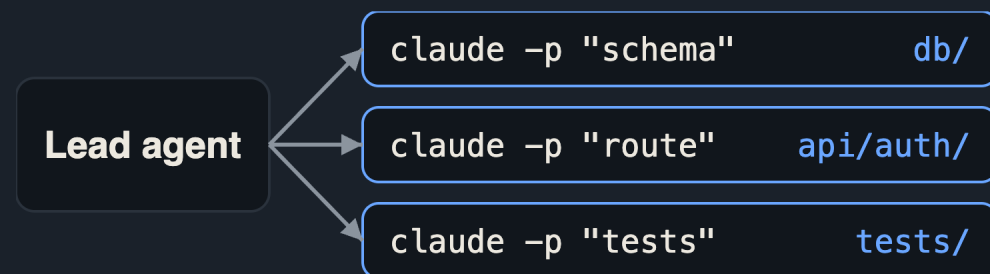
Give it bash and it writes its own tools. You can't predict what it will run, so run it in a sandbox.

Headless execution

```
$ claude -p "Fix the failing test in auth/" --output-format json
```

Decompose

Split a big task. Each subagent gets one job and only the context it needs.

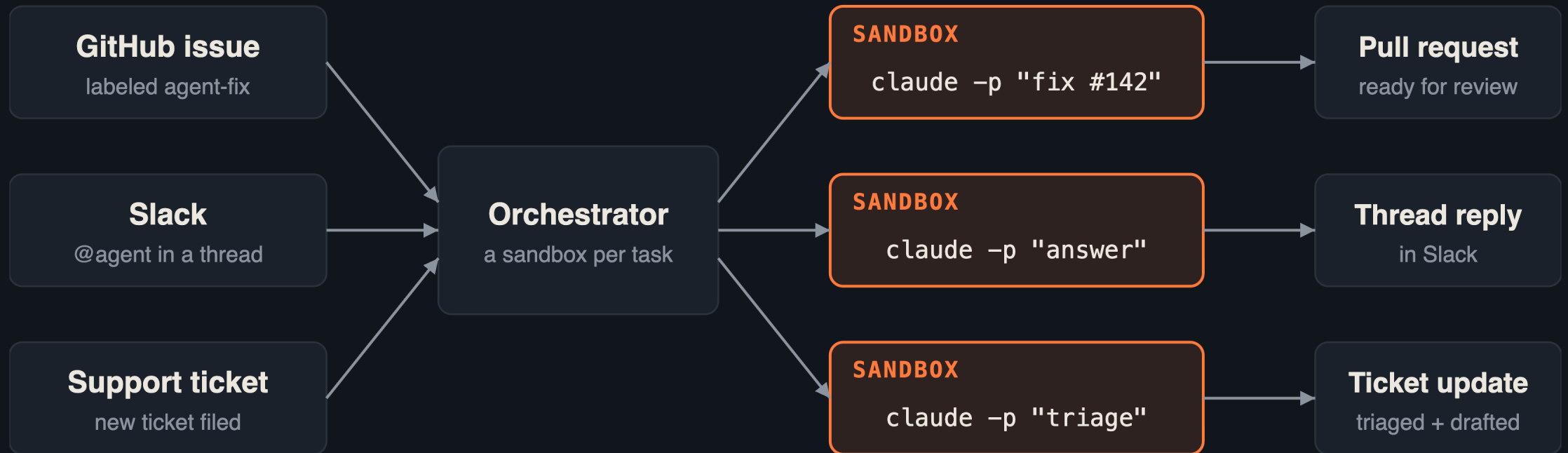


Automate

No one at the keyboard. Events and schedules start the agent.



Orchestrate



Same agent, many entry points. **Every task gets its own sandbox.**

STAGE 1 OF 5 • ON YOUR LAPTOP

Autocomplete

LOOKS LIKE

AI suggests. You type,
run and check
everything.

WHAT BREAKS

You want the AI to run
the code, not just write it.

BOTTLENECK

Your typing

1 Autocomplete

2 Local agent

3 Parallel local

4 Agent factory

5 Agent swarm

Local agent

LOOKS LIKE

An agent in your terminal runs commands. OS-level sandboxing lets you relax the prompts.

WHAT BREAKS

You spend the day clicking Allow.

BOTTLENECK

Your approvals

Parallel local agents

LOOKS LIKE

Several agents
in worktrees and
devcontainers, in
YOLO mode.

WHAT BREAKS

Ports, RAM, credentials
and blast radius, all on
one machine.

BOTTLENECK

Your hardware

Agent swarm

LOOKS LIKE

Many agents on one problem: fan out, fork, compare.

WHAT BREAKS

Cold start and cost per sandbox become architecture decisions.

BOTTLENECK

**Isolation vs.
speed vs. cost**

What a sandbox is

An ephemeral, isolated computer with an API, where an agent can run anything inside a blast radius **you control.**

Isolated

Ephemeral

Programmable

Fast to start

Snapshottable

The isolation spectrum

OS sandbox

Seatbelt,
bubblewrap,
Landlock

Container

Namespaces and
cgroups; shared
host kernel

gVisor

User-space kernel
intercepts syscalls

microVM

Firecracker, Kata;
own kernel

Full VM

Full hardware
virtualization

← FASTER START • LESS OVERHEAD

STRONGER BOUNDARY →

Agent factory

LOOKS LIKE

Tickets in, PRs out. One disposable sandbox per task.

WHAT BREAKS

One attempt per task isn't enough for hard problems.

BOTTLENECK

Review & verification

Who builds the sandboxes

Tool	Isolation	Notable
Firecracker	microVM	Open-source building block from AWS; not a product
Docker	Container	Dev default; not enough alone for untrusted code
E2B	Firecracker microVM	Agent-focused code-interpreter SDK
Modal	gVisor	Built for scale; locked down by default
Daytona	Container (Sysbox)	Fast creation; egress allowlists; placeholder secrets

Laptops are where agents learn to code.
Sandboxes are where they learn to scale.

Thank you · Questions?

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